Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 128. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the test accuracy during training.

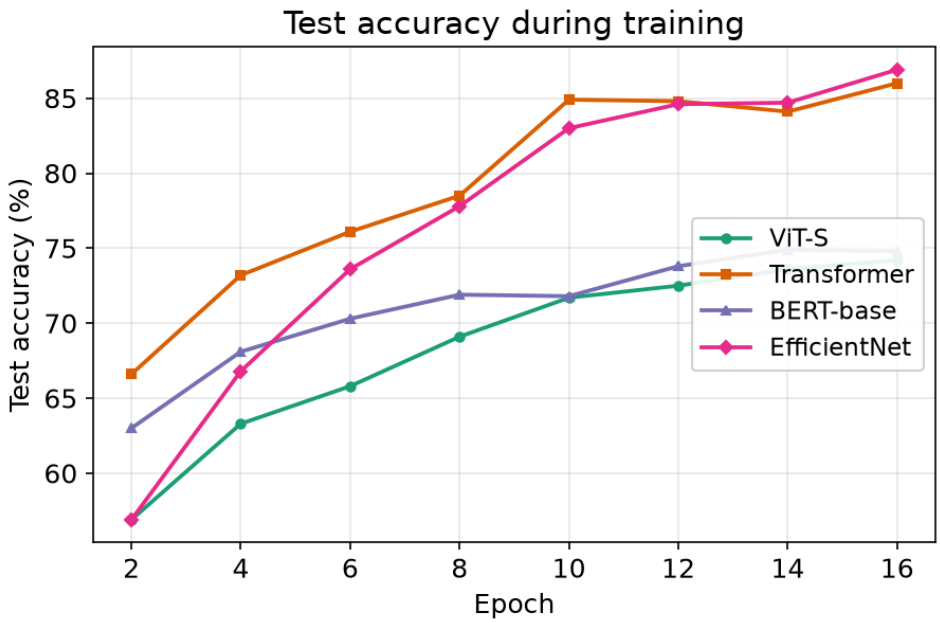


Figure 1: Test accuracy during training for the models in this experiment.

As shown in Figure 1, at epoch 8, EfficientNet is 5.4 points better than ViT-S. From epoch 4 to epoch 6, the test accuracy of ViT-S improves by a relative 3.9%. Transformer attains a test accuracy of 68.3% at epoch 12. As shown in Figure 1, Transformer reaches 67.6% test accuracy at epoch 14.

Figure 2 shows the test accuracy by dataset.

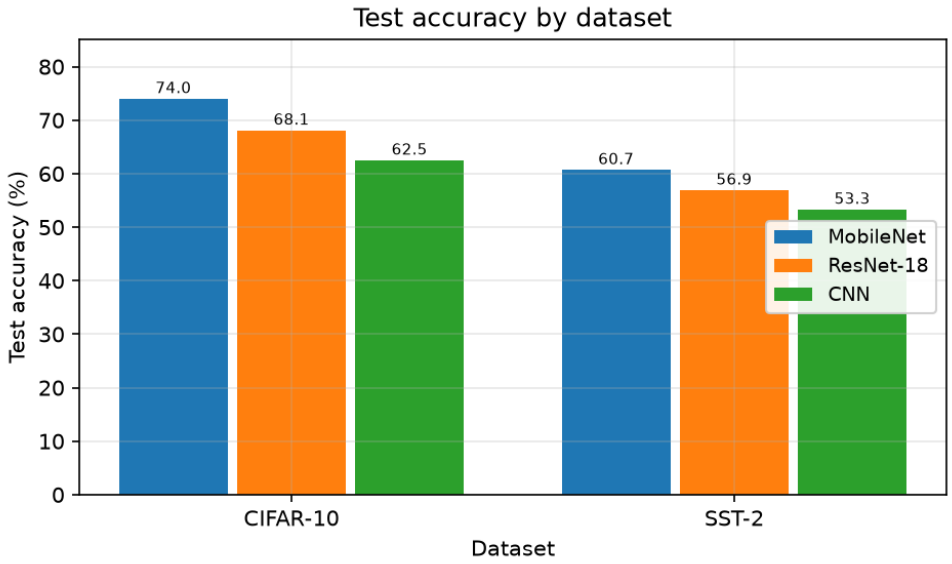


Figure 2: Test accuracy by dataset for the models in this experiment.

On SST-2, MobileNet is 46.4 points better than ResNet-18. MobileNet outperforms ResNet-18 by about 6 percentage points on CIFAR-10 (Figure 2). On SST-2, CNN obtains a test accuracy of 95.8% (Figure 2).

Figure 3 shows the test accuracy of CNN by dataset.

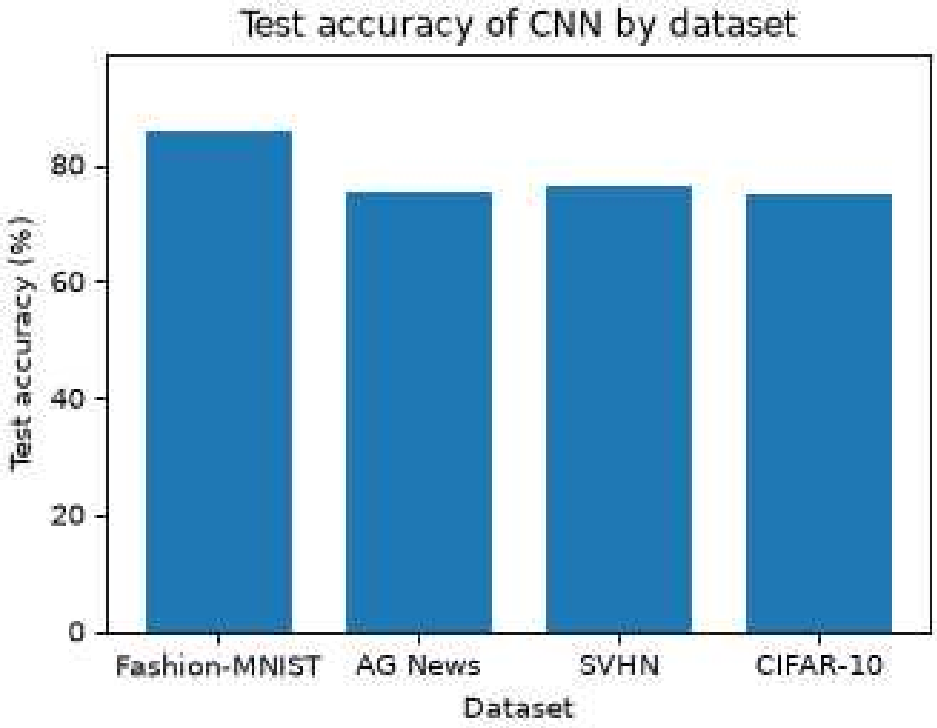


Figure 3: Test accuracy of CNN by dataset for the models in this experiment.

CNN scores 12.5 points higher on Fashion-MNIST than on AG News (Figure 3). CNN achieves 55.6% test accuracy on AG News (Figure 3).

Figure 4 shows the f1 score during training.

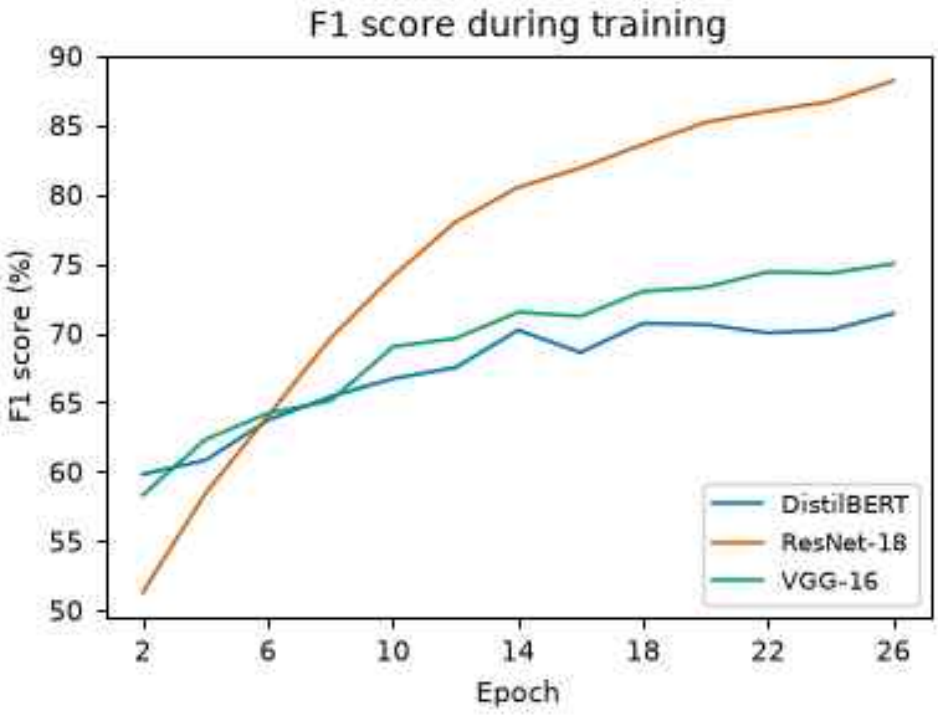


Figure 4: F1 score during training for the models in this experiment.

At epoch 14, ResNet-18 achieves 84.6% F1 score. At epoch 20, VGG-16 achieves 73.3% F1 score (Figure 4). DistilBERT reaches 65% F1 score at epoch 8. At epoch 12, VGG-16 is 2.1 points better than DistilBERT (Figure 4).

3 Discussion

We note that the training was stable for all models.